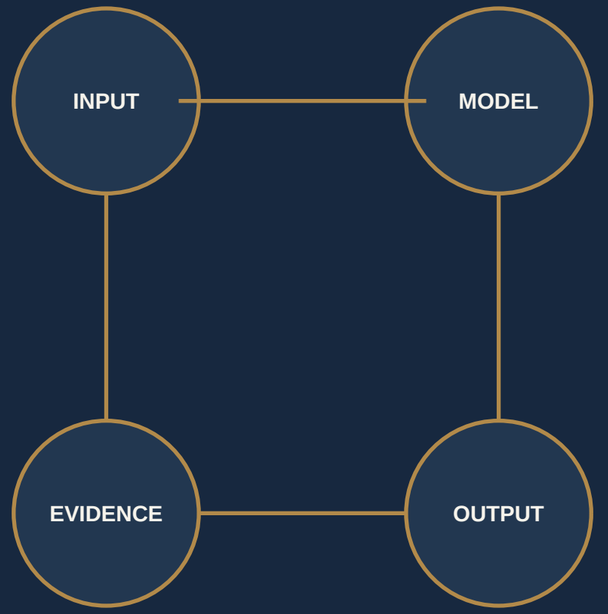
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robotics Planning

AI.311 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define state-space planning.
- Apply key methods including search.
- Build or analyze a practical system involving task planning.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

State-space planning

02

METHODS

Search

03

SYSTEMS

Task planning

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · State-space planning

State-space planning

02 · Search

Search

03 · Task planning

Task planning

04 · Motion planning

Motion planning

05 · Trajectory generation and ma

Trajectory generation and manipulation planning

06 · Mandatory practical: Implement

Mandatory practical: Implement planning algorithms and generate collision-free robot motions

Core ideas, made explicit

AI.311 • Distinguish definitions, assumptions and evidence

- State-space planning
- Search
- Task planning
- Motion planning
- Trajectory generation and manipulation planning

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.311 • Distinguish definitions, assumptions and evidence

- Mandatory practical: Implement planning algorithms and generate collision-free robot motions

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
State-space planning

INPUTS
Search

METHOD
Task planning

OUTPUT
Motion planning

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving state-space planning.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using state-space planning.
- Implement or simulate a baseline using search.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.